



An Empirical Ethics Scale for Technology Assessment: Challenges and Perspectives for a Real Time Ethics

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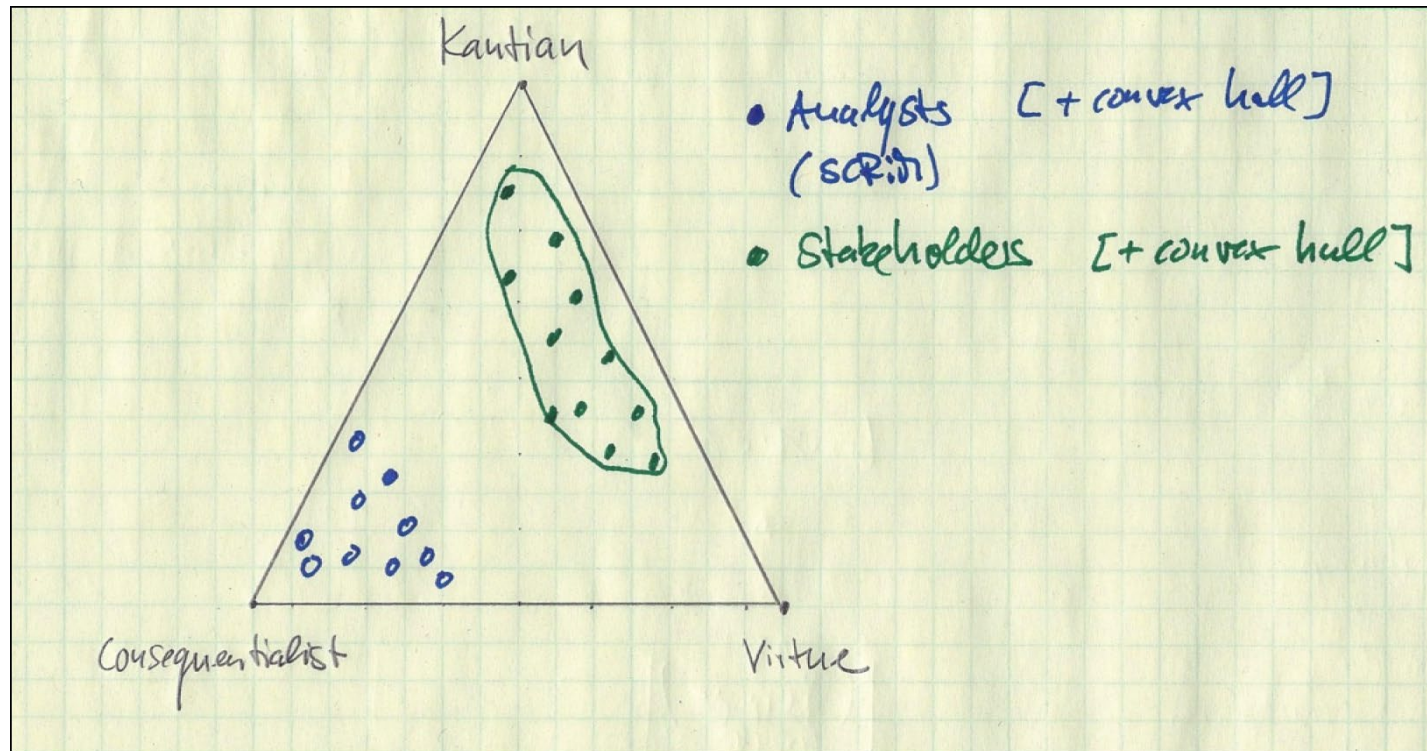
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Structure of our presentation:

- (1) Challenges for technology assessment and flood risk management: stakeholder attitudes
- (2) Responses: inter- and transdisciplinarity, empirical ethics, and real-time ethics
- (3) Methodological implementation: ethics scale on six key ethics types
- (4) Validation of developed scale
- (5) Statistical opportunities
- (6) Conclusions

(1) Challenges for technology assessment and flood risk management: stakeholder attitudes



Slide by Parker and Tuana, NCAR

- **Great Acceleration** in the Anthropocene (accelerating technological development, waste, carbon dioxide,...)
- **Emerging technologies** with normative uncertainties
- **Ethical decisions** are required that involve various **stakeholder groups** as well as the perspectives of **future generations**

(2) Responses: Inter- and transdisciplinarity, empirical ethics, and real-time ethics

(a) Empirical ethics

- cooperation of moral psychology and ethics
- descriptive and normative
- description, interpretation, explanation, evaluation and change of social practices by using empirical data
- identification of external conditions that influence the process of ethical decision making

(b) Inter- and transdisciplinarity

- involvement of various societal groups and sciences to research in emerging technologies
- interdisciplinary evaluation at the technology level, the artifact level and the application level
- Incorporation of transdisciplinary perspectives (perspectives from various social groups)

(c) Real-time ethics

- technology assessment at an early stage is often based on insufficient information (possible side-effects on society and nature)
- “Collingridge dilemma” (inverse proportionality between sufficient information and possibilities for revision)
- reduction of the normative uncertainty of emerging technologies

(3.1) Methodological implementation: ethics scale on six key ethics types

Questionnaire to determine the test persons' ethical judgement of a technology

Most validated Ethics Scales:

- Multidimensional Ethics Scale by Reidenbach & Robin 1988, 1990

Shortcomings in terms of content and psychometrics

- central moral theories are missing
- based on too few items
- Background in business ethics

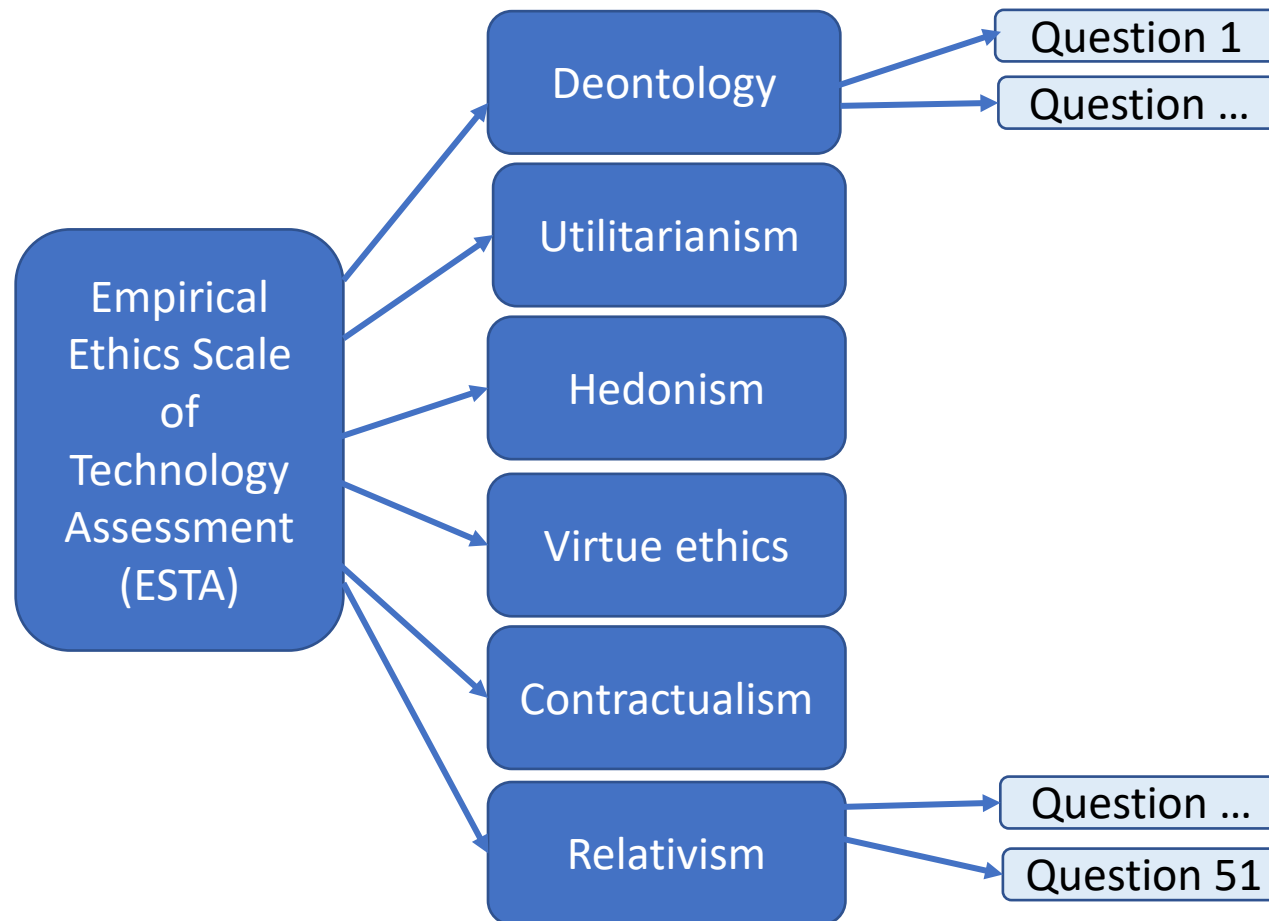
→ limitations in the context of technology assessment

Benefits of an Empirical Ethics Scale for Technology Assessment

- Ethical assessment of a technology in terms of **several moral principles**
- Online studies reaching out to people with
 - a) *no* expertise, b) *interactional* expertise and c) *contributory* expertise
- **Inclusion of public perspectives and values** brings ethical conflicts between groups of actors to the surface

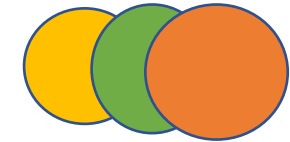
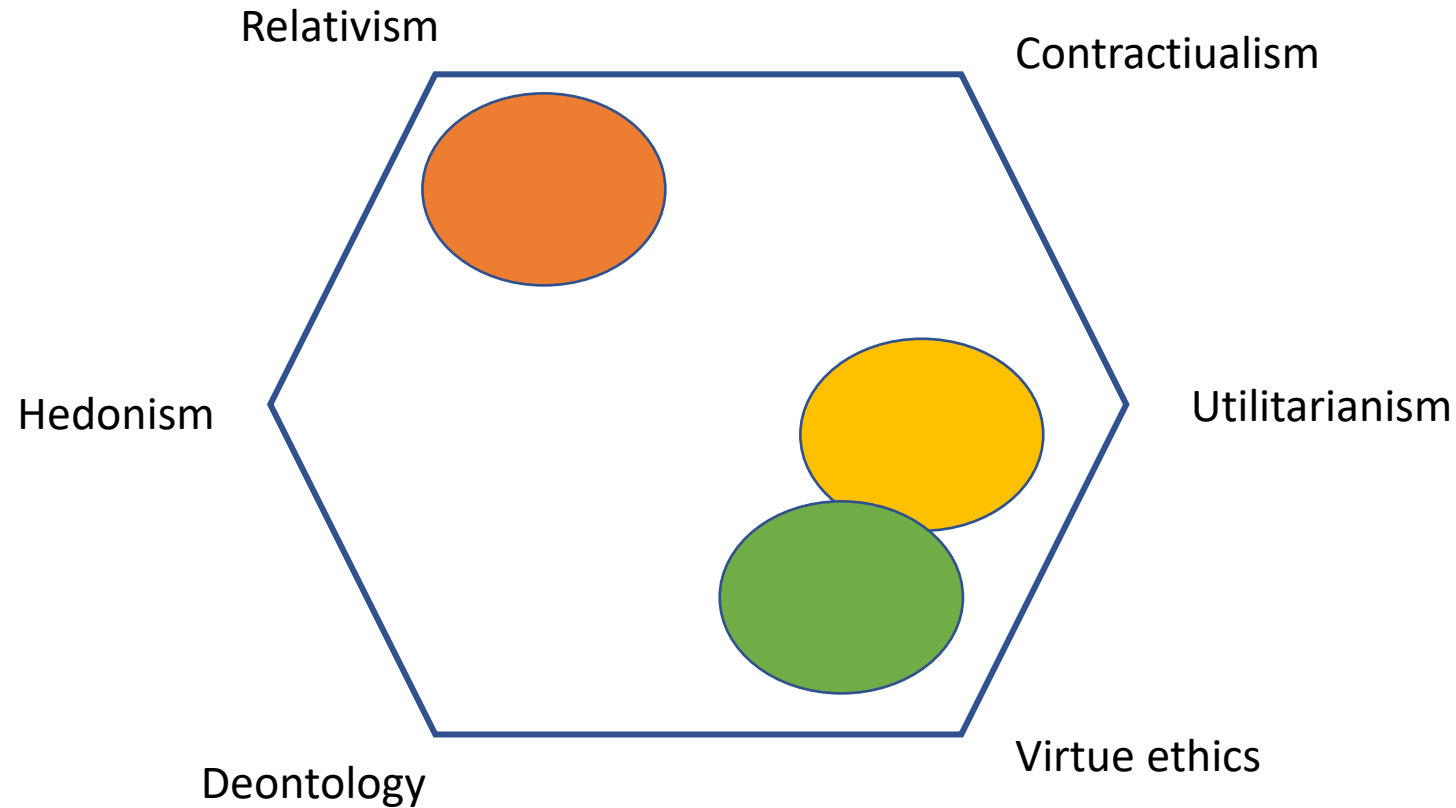
(3.2) Methodological implementation: ethics scale on six key ethics types

Ethical input in the form of questions on six key types of ethics (attention is directed in a specific direction by experts) and at the same time **integration of the societal perspectives of ethical laypersons**:



Deontology	A technology is morally right if it respects basic principles and duties.
Utilitarianism	A technology is morally right if it leads to the greatest good for the greatest number.
Hedonism	A technology is morally right if it increases pleasure, well-being or promotes a good life of an individual agent.
Virtue ethics	The developer is a genuinely virtuous person and has developed the technology taking into account her virtuous characteristics.
Contractualism	A technology is morally right if it does not violate moral norms that arise from mutual consensus.
Relativism	The moral judgement of a technology depends on the personal, societal or cultural contexts, which influence the standards of justification and procedures of justification.

(3.3) Methodological implementation: ethics scale on six key ethics types

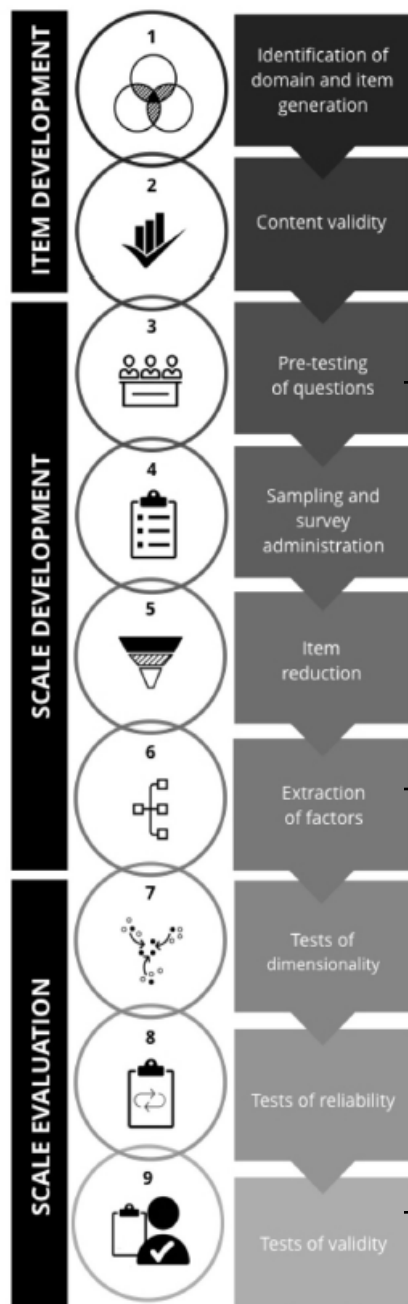


- different stakeholder groups
- different age
- different origin
- different levels of education
- ...

Fenn, J.*, Höfele P.*, Livanec S., Reuter, L., Müller, O. & Kiesel, A. (in prep.). An Empirical Ethics Scale for Technology Assessment: Challenges and Perspectives for a Real Time Ethics for Emerging Technologies.

(4) Forthcoming: validation study

Validation of the “Ethics Scale of Technology Assessment”



Study 1: Item generation and item review
> Pretest developed items qualitatively

Study 2: Examining dimensionality
> Get dimensionality of scale, scale purification

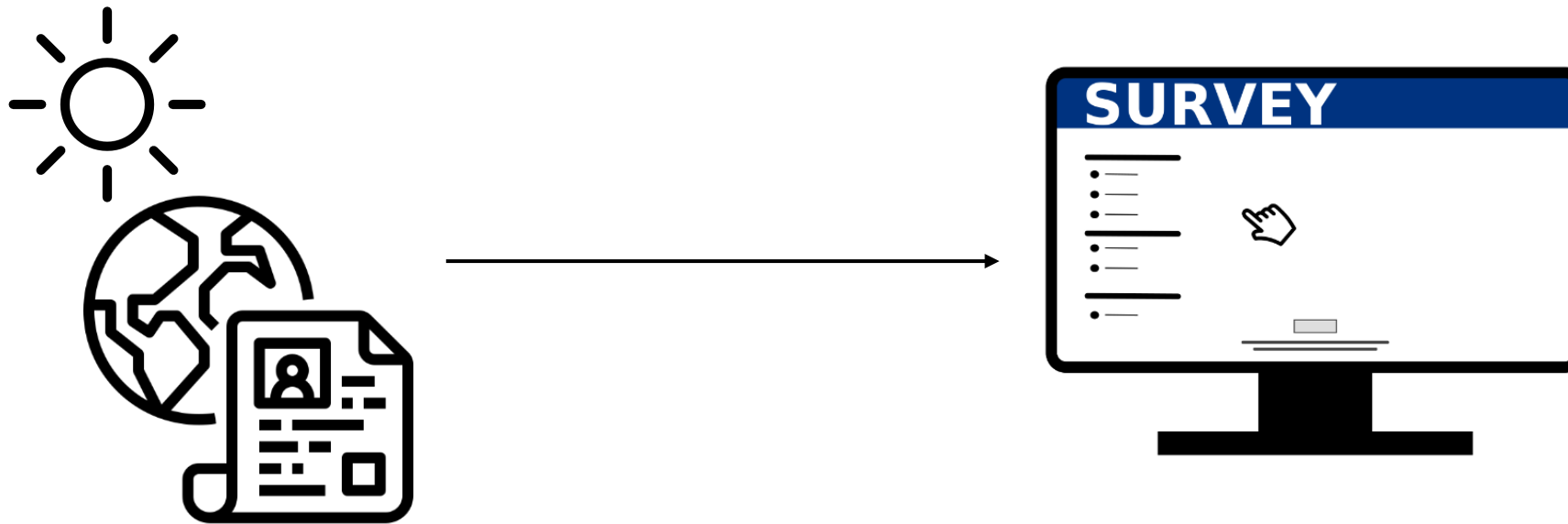
Study 3: Examining the *nomological net*
> Determine construct validity

Fenn, J., Höfele, P., Gros, W., Helm, J., Müller, O. & Kiesel, A. (in prep.). The Ethics Scale of Technology Assessment: Validation of a Measure for the Empirical Assessment of Emerging Technologies.

(5) Statistical opportunities: collecting data

Possible study design:

- 1) A scenario-text is presented to the participants
- 2) Afterwards they will answer the “Ethics Scale of Technology Assessment” (and additional questions)



(5) Statistical opportunities: latent variable models

Reflexive measurement model assumes that the answer to the questions of a construct η (like Hedonism) are considered to be caused by that construct

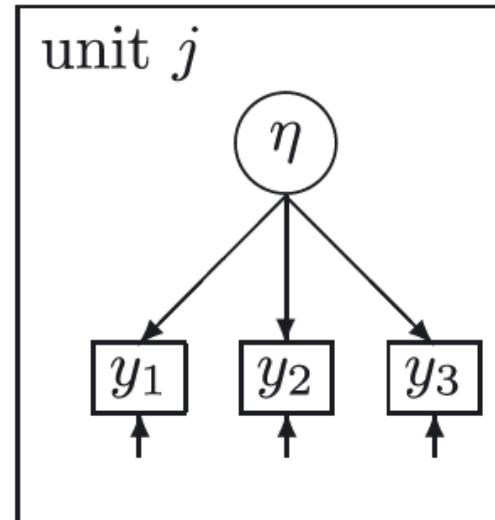


Figure 1 from Skrondal & Rabe-Hesketh, 2007

	Latent variable(s)	
Observed variable(s)	Continuous	Categorical
Continuous	Common factor model Structural equation model Linear mixed model Covariate measurement error model	Latent profile model
Categorical	Latent trait model/IRT	Latent class model

Table 1 from Skrondal & Rabe-Hesketh, 2007

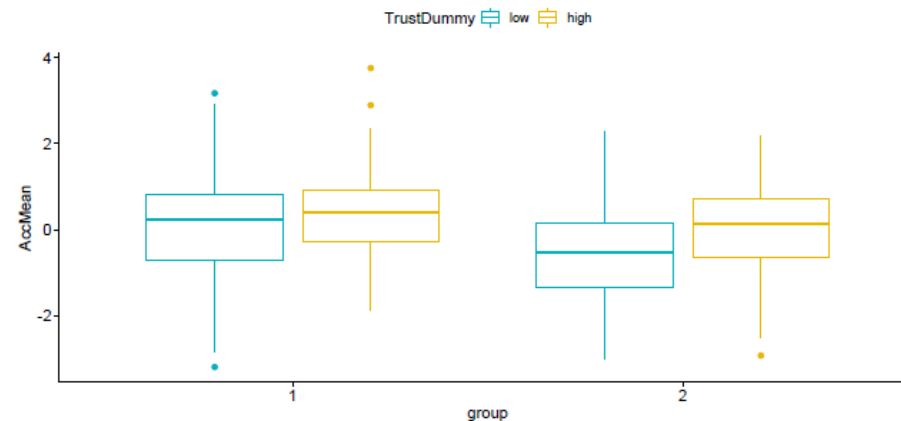
> we can apply the framework of *latent variable models*

(5.1) mean comparison

Are there significant mean differences in the ethical evaluation of technologies between laypersons vs. experts?

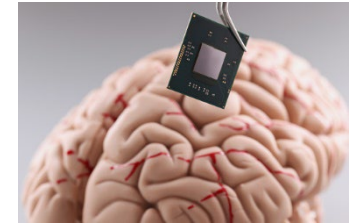
Test for mean differences* between different stakeholder groups:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	1	41.88	41.88	36.45	0
TrustDummy	1	23.52	23.52	20.47	0
group:TrustDummy	1	4.00	4.00	3.48	0.06
Residuals	496	569.92	1.15		



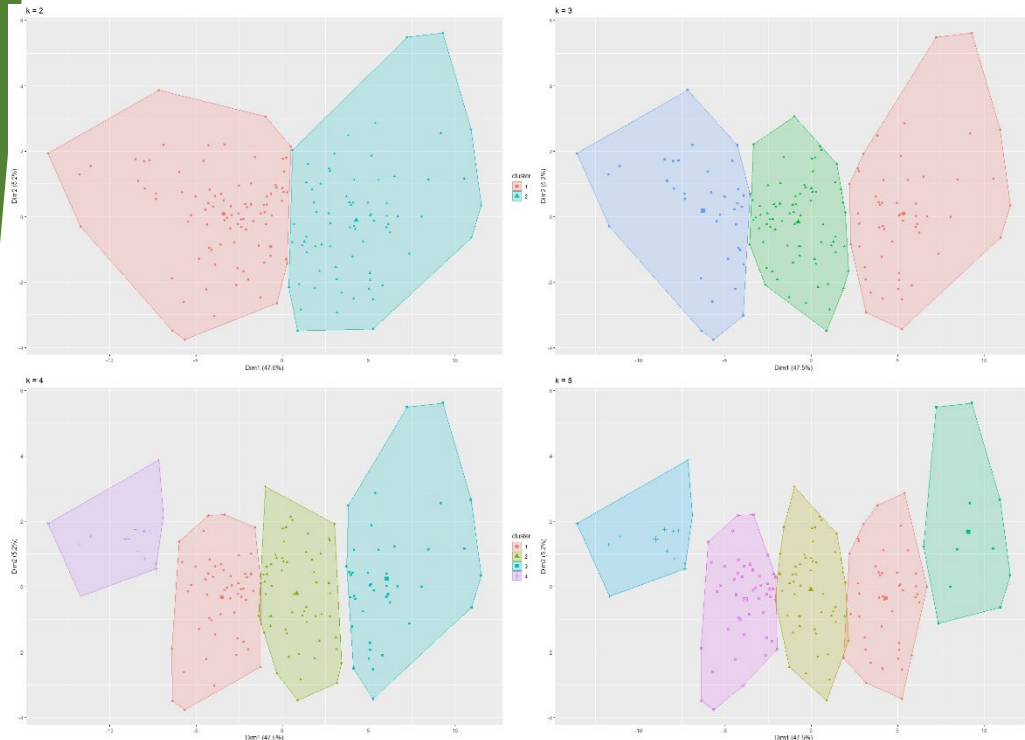
* precondition: (partial) scalar measurement invariance

(5.2) latent class analysis / cluster analysis



Are technologies ethically evaluated differently by participants (or laypersons vs. experts)?

Apply latent class analysis or cluster analysis (like k-means clustering)

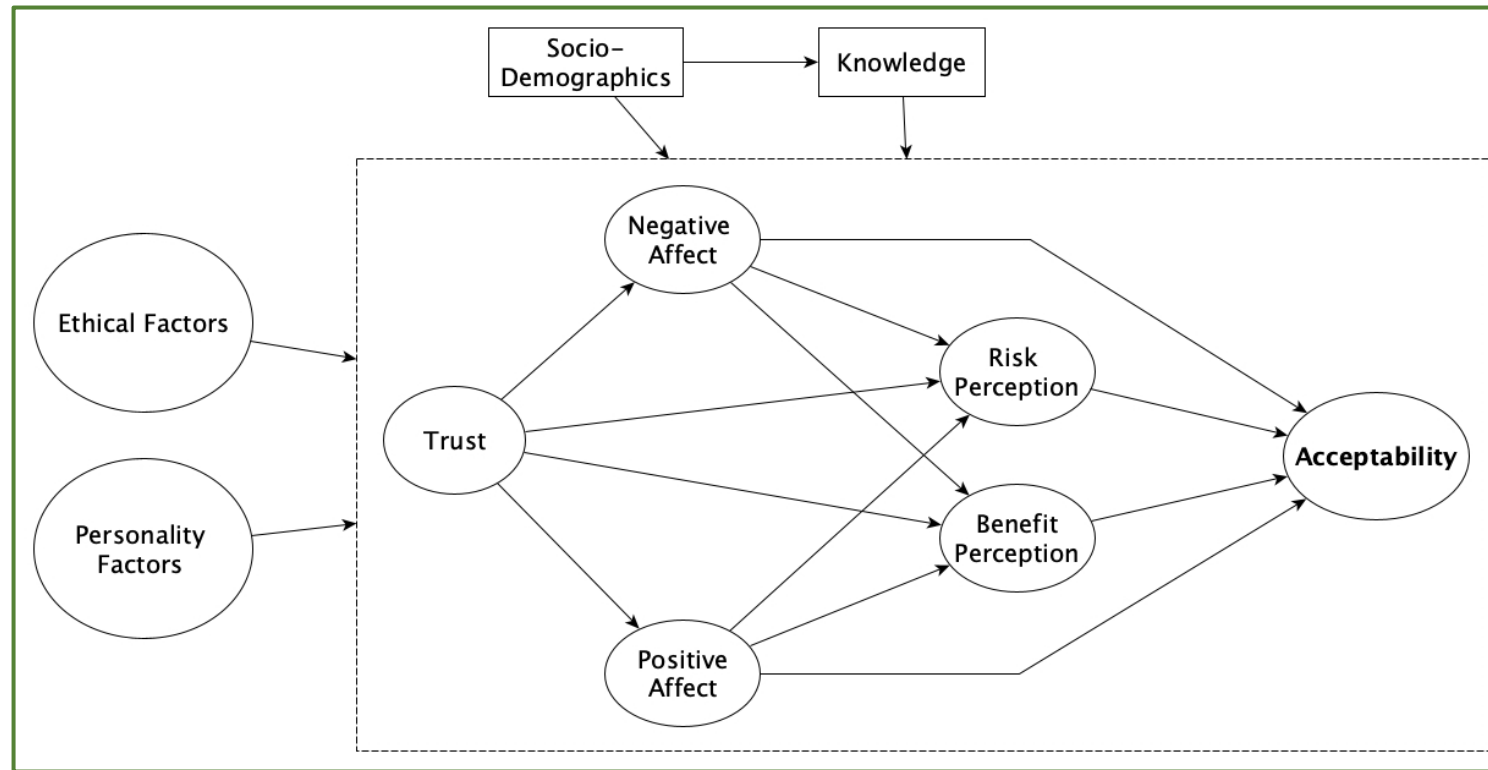


Cluster	Relativism	Contractualism	Utilitarianism	Deontology	Virtue ethics	Hedonism
1	2.43	2.58	3.32	2.79	3.14	2.94
2	3.79	4	4.71	3.89	4.36	4.2
3	5.22	5.29	5.77	5.01	5.43	5.39

(5.3) structural equation model

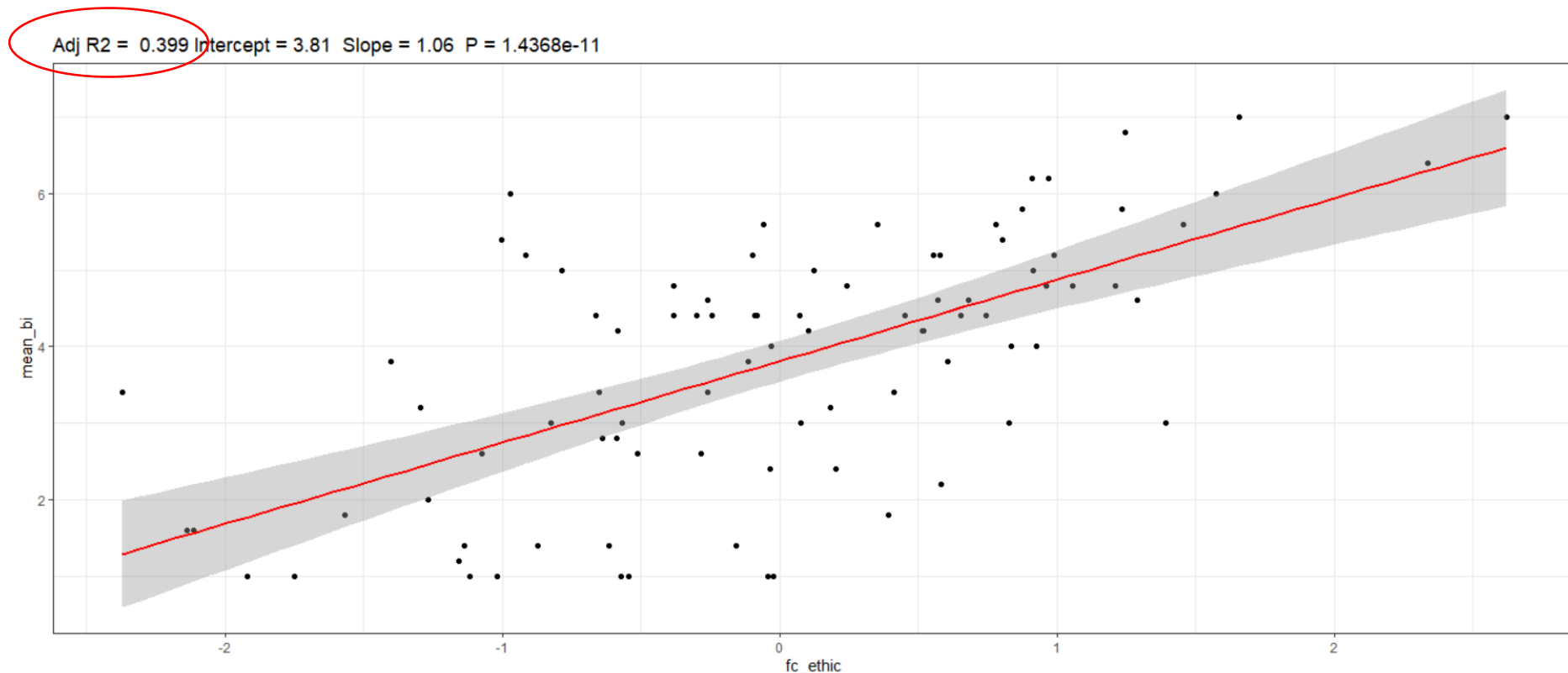
What is the effect of the ethical evaluation of a technology on its acceptability?

Test the direct and indirect effects on pre-assumed causal relationships:



(5.3) *ethic predict acceptability*

We conceptualize ESTA as *the degree respondents believe a technology is morally acceptable regarding multiple ethical theories*



(6) Conclusions

- The “Ethics Scale of Technology Assessment” allows a highly economic assessment of possible ethical concerns of different stakeholders
 - only represents a “**broad early warning system**”, needs to be supplemented by decision analysis, impact analysis or risk assessment, ... (Keller et al., 2021)
- To the best of our knowledge it is the first time possible to test if technologies are evaluated ethically differently
 - like iPhones are evaluated more strongly on hedonistic aspects (“the iPhone maximizes my pleasure”) vs. nuclear power plants are evaluated more strongly in a utilitarian perspective (“the nuclear power plant produces more benefits than harm for society”)
- Preliminary results showing the importance of the ethical evaluation on acceptability

Thank you for your attention!

Questions?



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Ethic Scale I

Ethics	Item number	negative (left)	positive (right)
Relativism	"relativist01"	"...is culturally unacceptable"	"...is culturally acceptable"
	"relativist02"	"...is unacceptable to my family"	"...is acceptable to my family"
	"relativist03"	"...is traditionally unacceptable"	"...is traditionally acceptable"
	"relativist04"	"...is individually unacceptable"	"...is individually acceptable"
	"relativist05"	"...is unacceptable to people I most admire"	"...is acceptable to people I most admire"
Contractualism	"contractualist01"	"...is unjust"	"...is just"
	"contractualist02"	"...is unfair"	"...is fair"
	"contractualist03"	"...respects an unwritten contract"	"...does not respect an unwritten contract"
	"contractualist04"	"...respects an unspoken promise"	"...does not respect an unspoken promise"
	"contractualist05"	"...does not result in an equal distribution of good and bad"	"...results in an equal distribution of good and bad"
	"contractualist06"	"...violates my ideas of fairness"	"...does not violate my ideas of fairness"
	"contractualist07"	"...impedes political rights"	"...supports political rights"

Ethic Scale II

Ethics	Item number	negative (left)	positive (right)
Hedonism	"hedonism01"	"...is personally unsatisfactory "	"...is personally satisfactory"
	"hedonism02"	"...would be selfish for me to use"	"...would not be selfish for me to use"
	"hedonism03"	"...requires me to make sacrifices in order to use it"	"...does not require me to make sacrifices in order to use it"
	"hedonism04"	"...is not in the best interests of my person"	"...is in the best interests of my person"
	"hedonism05"	"...minimizes my pleasure"	"...maximizes my pleasure"
	"hedonism06"	"...is a hindrance to a good personal life"	"...promotes a good personal life"
	"hedonism07"	"...harms my health"	"...promotes my health"
	"hedonism08"	"...harms my freedom"	"...promotes my freedom"
	"hedonism09"	"...harms my safety"	"...promotes my safety"
Utilitarianism	"utilitarian01"	"...provides the least utility for society"	"...provides the greatest utility for society"
	"utilitarian02"	"...minimizes benefits for society"	"...maximizes benefits for society"
	"utilitarian03"	"...maximizes harm for society"	"...minimizes harm for society"
	"utilitarian04"	"...tends overall to be bad for society"	"...tends overall to be good for society"
	"utilitarian05"	"...is not OK if actions can be justified by their consequences for society"	"...is OK if actions can be justified by their consequences for society"
	"utilitarian06"	"...leads to the least good for the greatest number"	"...leads to the greatest good for the greatest number"
	"utilitarian07"	"...leads to the greatest ill for the greatest number"	"...leads to the least ill for the greatest number"
	"utilitarian08"	"...results in a negative cost-benefit ratio for society"	"...results in a positive cost-benefit ratio for society"
	"utilitarian09"	"...is inefficient for society"	"...is efficient for society"
	"utilitarian10"	"...leads to future harm for society"	"...provides for future benefit society"

Ethic Scale III

Ethics	Item number	negative (left)	positive (right)
Deontology	"deontology01"	"...does not imply a moral obligation to act in a certain way"	"...implies a moral obligation to act in a certain way"
	"deontology02"	"...harms the autonomy of users"	"...promotes the autonomy of users"
	"deontology03"	"...obliges a certain immoral behaviour"	"...obliges a certain moral behaviour"
	"deontology04"	"...is not morally right"	"...is morally right"
	"deontology05"	"...goes against an important moral rule by which I live"	"...does not go against an important moral rule by which I live"
	"deontology06"	"...implies a moral obligation to act otherwise"	"...implies no moral obligation to act otherwise"
	"deontology07"	"...prevents treatment with respect in interactions"	"...ensures treatment with respect in interactions"
	"deontology08"	"...prevents treatment with dignity in interactions"	"...ensures treatment with dignity in interactions"
	"deontology09"	"...attacks the intrinsic value of nature"	"...protects the intrinsic value of nature"
	"deontology10"	"...attacks the value of the ecological environment"	"...protects the value of the ecological environment"
	"deontology11"	"...attacks the value of the cultural environment"	"...protects the value of the cultural environment"

Ethic Scale IV

Ethics	Item number	negative (left)	positive (right)
Virtue ethics	"virtue01"	"...is developed by someone who has wrong motivations"	"...is developed by someone who has good motivations"
	"virtue02"	"...is developed by someone who has wrong desires"	"...is developed by someone who has good desires"
	"virtue03"	"...is developed by someone who has a bad character"	"...is developed by someone who has a good character"
	"virtue04"	"...is developed by someone who is not prudent"	"...is developed by someone who is prudent"
	"virtue05"	"...is developed by someone who is not reasonable"	"...is developed by someone who is reasonable"
	"virtue06"	"...is developed by someone who is not striving for professional excellence"	"...is developed by someone who is striving for professional excellence"
	"virtue07"	"...is developed by someone who is indifferent towards nature"	"...is developed by someone who is respectful towards nature"
	"virtue08"	"...is developed by someone who is insensitive to interactions with society"	"...is developed by someone who is sensitive to interactions with society"
	"virtue09"	"...is developed by someone who is acting on ill intentions"	"...is developed by someone who is acting on good intentions"

